HIT140_A3_Group_Report.pdf

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HIT140 - Group Assessment : Predicting FIFA World Cup 2026: goal difference and goals scored with multiple linear regression

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1. Introduction

For this assignment, we analyze the World Cup 2026 by building two Linear regression models for 104 matches: LR 2.1 and LR 2.2. LR 2.1 predicts the goal difference of a match (104 rows of data), and LR 2.2 predicts the goal scored by a single team in a match (208 rows of data). Each regression model uses exactly 8 explanatory variables that are known before kick-off, and the two models share exactly 4 variables, so together, we use a total of 12 different variables. We also compare the final result with other regression models to see if simple Linear Regression is enough

2. Data sources and why we used more than the recommended three

In total, 5 datasets are used (Figure 1):

- **FBref** gives us the result and schedule, but not any information before the kick-off.
- **Elo ratings** estimate the strength of a team before each match.
- **Transfermarkt squad value** gives the estimated money value of a team in euros.
- **World Cup history** adds to each nation's experiences.
- **2018 and 2022 World Cup statistics** (possession, shots, corners, cards, fouls, clean sheets) show the statistics of each team before kick-off.

We could not use FIFA rankings as the pages we found were published after the tournament and would leak results, and we did not use any 2026 in-match statistics.



Figure 1. files are cleaned, joined with one team spelling, and turned into leak-safe variables.

Figure 1 is an illustration of how data flows through the project. There are five datasets which are inputs to this flow: 2026 games on FBref, Elo ratings, squad valuation from Transfermarkt, World Cup history, and match stats from the years 2018 and 2022. These five inputs are cleaned up and merged with the same spelling of the teams and variables created using only pre-game data. These variables are used as inputs to two regression models, namely LR 2.1 (104 observations) and LR 2.2 (208 observations) with 4 common and 4 different variables respectively.

3. How we chose the variables

Availability audit:

We checked every existing variable and kept only those that existed before the match. Some variables contain the result, such as Elo of a team after a match. We also include some historical variables, such as historical possession (hist_poss). These variables are calculated from the 2018 and 2022 World Cup datasets by using a shrunk average, (statistics so far + 2 x tournament average)/ (matches so far +2), instead of zeros. Not all teams that participated in the 2026 World Cup were competing in the 2018 and 2022 World Cups so for them, we used average values of all teams and since it varies and takes all teams, strong or weak over different editions, it is okay to do so. We also understand that we cannot take the average of sensitive target variable or feature variables such as scores, assists, goals and so on.

EDA selection:

After getting a list of candidate variables, we decided the expected sign of the variable correlated with the result variable. For example, we expected that the higher the elo rating difference between two teams is, the bigger the goal differences. Therefore, we assign it a positive sign. Some have 0 as we cannot predict how it affects the result variable. Next, we use Pearson correlation and VIF on the first 83 matches only (as we split 80% of the dataset to allocate for training and remaining 20% for testing (section 4) which is splitted in time based order) and choose the best 12 variables [8 unique and 4 shared among them]. Four strongest variables on average across both tasks are shared, and 4 more unique variables are chosen for each task. All 16 chosen terms had the expected sign (correlation) and every VIF was below 5.

Type	LR 2.1 goal difference (home - away)	r	LR 2.2 goals scored (team)	r
Shared (4)	Squad value (log) difference	0.69	Squad value (log) difference	0.55
	Elo before match, difference	0.64	Elo before match, difference	0.51
	Earlier World Cups played, difference	0.55	Earlier World Cups played, difference	0.46
	Historical World Cup goal diff. per match, difference	0.34	Historical World Cup goal diff. per match, difference	0.29
Unique (4)	Possession 2018/22, difference	0.32	Own shots per match 2018/22	0.25

	Confederation strength (UEFA/CONMEBOL), difference	0.28	Opponent clean-sheet rate in 2026 so far	-0.23
	Corners 2018/22, difference	0.23	Opponent goals conceded per game in 2026 so far	0.21
	Fouls 2018/22, difference	-0.14	Opp. shots on target conceded 2018/22	0.13

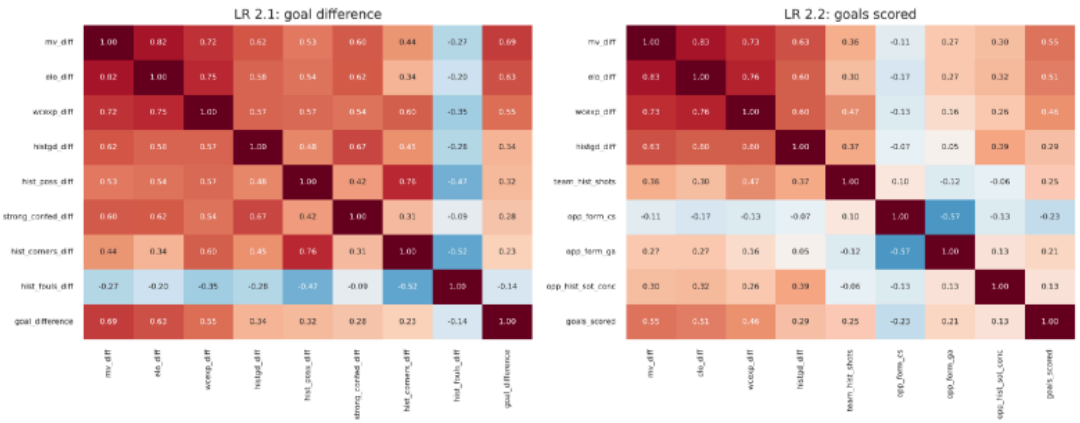


Figure 2. Correlations among the chosen variables and the target (training matches only). The strength variables are linked with each other, which explains the unstable single coefficients later.

Figure 2 is a visual illustration of the correlations between the selected variables and the target based on the training set of 83 matches only. The strength-based variables (team value, Elo rating, and World Cup experience) are positively correlated, with a correlation coefficient between Elo rating and team value being around 0.82. This strong correlation makes the coefficients unstable in further analysis. Speaking about the strength of correlation with the target, the team value has a correlation of 0.69 in LR 2.1 and 0.55 in LR 2.2, while Elo rating has 0.64 and 0.51, which makes them the best predictors. The style-based variables, on the contrary, have weaker correlations of about 0.13 – 0.32.

4. Method

Matches were split in 80% for training and 20% for testing so we have 83 matches for training and 21 matches to test our model. After the initial linear regression, we perform 5 optimization steps to enhance the accuracy of the model:

- Outlier removal: for LR 2.1, there are 4 matches with a big goals difference (4 goals and above). However, removing the outlier makes the cross-validation of RMSE worse (from 1.559 to 1.616), so we decided not to remove the outlier. The opposite happened for LR 2.2, where RMSE decreased (from 1.228 to 1.181), so we kept it for LR 2.2 only

- Non-linearity check: After sketching all scatterplots for each variable, we see no non-linearity pattern, so nothing is optimized in this step.

- Multicollinearity check: highest VIF 4.8 and strongest correlation between 2 variables is 0.82 for LR 2.1 and highest VIF 4.0 and strongest correlation is 0.83 for LR 2.2, so no variable should be excluded.

- Yeo-Johnson transformation: we keep this transformation for LR 2.2 only as it did not improve the result for LR 2.1

- z-standardisation: same predictions, comparable coefficients, so we kept for both linear regression.

5. Results

	LR 2.1 OLS	Mean baseline	LR 2.2 OLS	Mean baseline
R^2 / adjusted R^2 (training)	0.554 / 0.506	-	0.329 / 0.292	-
Cross-validated RMSE (goals)	1.559	2.171	1.179	1.417
Test RMSE / MAE, 21 matches	1.469 / 1.162	1.747 / 1.426	1.273 / 0.913	1.322 / 1.045
Backtest RMSE, 56 matches	1.335	1.956	1.248	1.293

LR 2.1 (goal difference). Of the 83 games used in training, it accounts for 55% of goal difference variance (adjusted R^2 = 0.51). Of the 21 unseen games, the story looks less impressive: while the average error is 1.47 goals compared to 1.75 for simply predicting the mean, R^2 drops to 0.08. It correctly picked the winner of 14 out of 19 non-draws (74%), compared to 9 (47%) out of 19 if we always selected the team first listed on FBref. The most important predictor is squad valuation: on average, the side whose squad valuation exceeds the other team's by one standard deviation wins by about 1.3 goals ($p<0.001$; see Figure 3). Both Elo and World Cup experience contribute in the right direction, but their confidence intervals contain zero, since they correlate with squad valuation (Elo correlates with squad valuation at 0.82). Thus, we cannot either confirm or reject the relative importance of these predictors. Despite its positive correlation with goal difference (+0.28), confederation strength has a negative coefficient, which we interpret as the result of multicollinearity (once squad valuation and Elo are taken into account, a European or South American team is slightly less strong than predicted).

LR 2.2 (goals scored). However, predicting the goals of a specific team proves to be much more difficult. The model explains 33% of variance in the training data set (adjusted R^2 = 29%), but only 6% for the 21 games not included in the

sample, where the average difference is 1.27 goals, as opposed to 1.32 goals for the average model. Value of squad appears to be the only non-zero predictor ($\beta = +0.49$ goals per standard deviation, $p < 0.001$; standard errors were calculated taking into account the fact that each match appears twice in the sample).

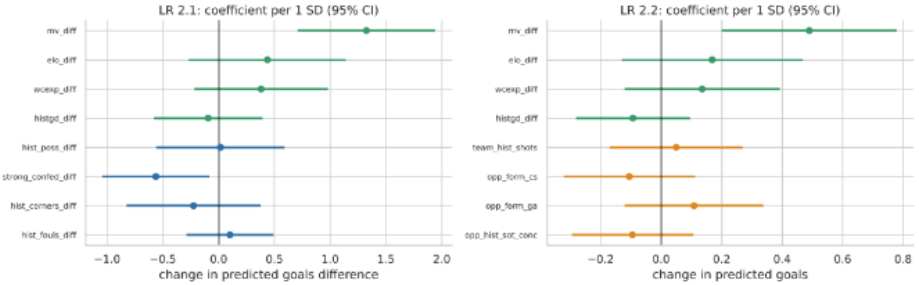


Figure 3. Standardised coefficients with 95% confidence intervals (green = shared variables).

In Figure 3, we can see the coefficient estimates for each variable, along with the 95% confidence intervals, scaled to one standard deviation. If the confidence interval also includes zero, then there is no observable effect of the variable from the data. In both regressions, the only variable that is definitively positive is the value of squad. The strength of confederation in regression LR 2.1 is the other variable whose interval does not include zero, but its sign is negative.

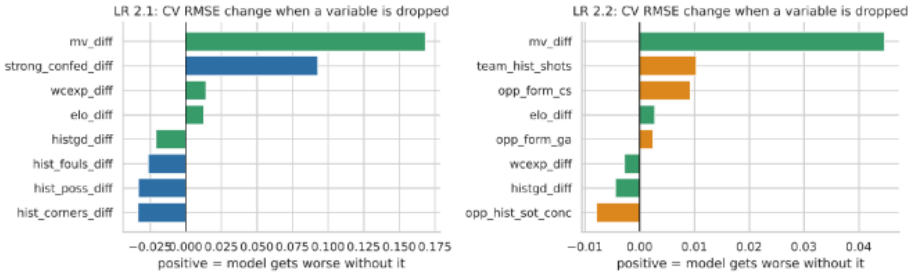


Figure 4. Change in CV error when one variable is dropped (green = shared). Dropping squad value hurts most; most unique variables matter little.

Figure 4 illustrates the change in cross-validated error due to the omission of each variable. A bar pointing to the right means an increase in error (a deterioration in model performance due to the omission of the particular variable), while a bar pointing to the left implies a marginal drop in error.

The omission of the squad value variable contributes to the greatest drop in performance for both models (with the error increasing by roughly 0.17 goals in LR 2.1 and 0.05 in LR 2.2). The omission of the confederation strength variable is the second biggest source of damage for the LR 2.1 model. On the contrary, the omission of the possession, corners, or fouls variables for LR 2.1, or the number of shots on target of the opposition in LR 2.2, leads to a marginal drop in error.

6. Competing models and recommendations

These models were compared using Ridge, Lasso, Elastic Net, kNN, SVR, Random Forest, Gradient Boosting, and for LR 2.2, Poisson regression on exactly the same sets of variables and splits. Settings were optimized based only on training matches. Figure 5 displays a backtesting error.

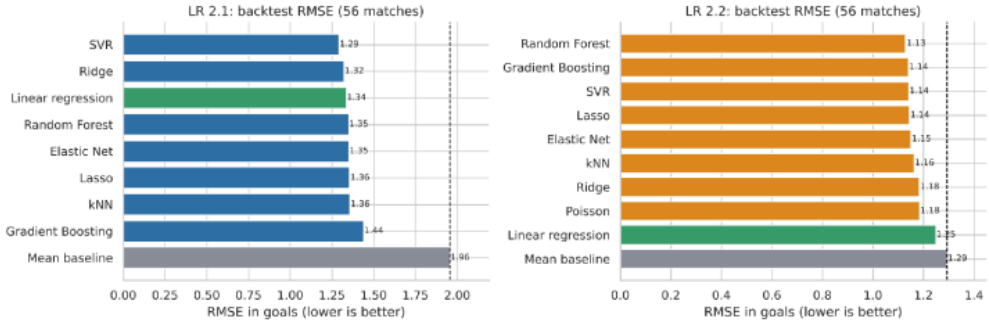


Figure 5. Backtest RMSE for all models (dashed line = mean-only baseline).

Figure 5 shows the backtest error for all models, with lower values reflecting better results. The dashed line is the baseline (mean-only), and the linear regression approach is shown in green. In LR 2.1, all models are significantly left to the baseline (varying between 1.29 and 1.44

compared with 1.96), but there is little difference between top-performing models. In LR 2.2, the model errors are grouped around the baseline (varying between 1.13 and 1.25 compared with 1.29). The minimum error belongs to the Random Forest, while linear regression is closest to the baseline.

LR 2.1. All the models used beat guessing the average by a big margin (backtest error 1.29 to 1.44 goals against 1.96). The differences between the ones that are best are quite small: SVR scores 1.29, Ridge 1.32 and plain linear regression 1.34, and the bootstrap says SVR's edge over linear regression is only +0.04 goals (95% range -0.13 to +0.21), which could easily be luck. In the 21 held-out matches SVR looks better (1.26 against 1.47), but it would not be a clear insight to lean on that with such few samples of data. A more complex model is not clearly worth it here.

LR 2.2. In the present case, linear regression is the worst performing model (-0.04 vs 1.29 of the average; the 95% confidence interval of linear regression, -0.06 to +0.17, covers zero). Random Forest wins the backtest (1.13; it beats linear regression by +0.12 goals, with the 95% CI of +0.02 to +0.21), followed by Gradient Boosting, SVR, Lasso and Elastic Net. Similarly, Random Forest wins the test on held out games (1.22 vs 1.27 of linear regression and 1.32 of the average).

What we recommend. Goal difference should be predicted using simple linear regression or Ridge since it is simple to understand and equally accurate as any other method. If it is important that the model should be interpretable, then goals scored can be predicted using either Lasso or Elastic Net. Otherwise, Random Forest should be used. **Two cautions:** Two warnings. First, the variables and parameters are selected using the data from matches 1 to 83, which means that the first part of the backtest (matches 49 to 83) is not entirely independent; the 21 matches that were used to check the model are relatively independent but not many of them. Second, we chose Random Forest since it was the best among a number of different models; therefore, the confidence interval for this method is slightly too narrow. There is a lot of luck in football, and no model can predict the score precisely.

7. Conclusion and limitations

Across both linear regression, squad market value was the most valuable predictor, while the historic variable contribute relatively little to the predictive value. It implies that pre-match team strength is useful, but it cannot fully capture the uncertainty of football.

Using external data such as Squad market value and Elo rating, we were able to create a decent linear regression for LR 2.1 that outperformed the mean baseline, with RMSE 1.34 and 1.96 respectively. Meanwhile, the linear regression for LR 2.2 does not provide much difference where the RMSE is 1.25 while the mean baseline RMSE is 1.29. This spot has a limitation that we might need other explanatory variables other than the one that we currently have to better predict the goals scored by a team. Overall, the findings support linear regression as an effective approach to predicting goal difference, while individual goal prediction remains a more challenging task that may benefit from alternative modelling techniques.

Please note: Code, data and all figures are in the repository linked in the submitted .txt file.

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